

SC1007 Data Structures and Algorithms

Lab 8: Trie

Practice Questions

Q1 Modify the `_add_child()` method so that it always inserts characters in an alphabetical order.

For example:

Insert: 'c', 'a', 'b'

LinkedList: a -> b -> c

Insert: 'e', 'd'

LinkedList: a -> b -> c -> d -> e

Q2 With the modified `_add_child()` method, implement a method to print all words stored in a trie in a lexicographical order.

```
def print_words_alphabetically():
```

For example, Insert: "banana", "apple", "bat", "ball", "band", "cat"

The results to print words in a lexicographical order: apple, ball, band, banana, bat, cat

Q3 Modify the method of Q2 to print all words stored in a trie in a reverse lexicographical order.

```
def print_words_reverse_alphabetically():
```

For example, Insert: "banana", "apple", "bat", "ball", "band", "cat"

The results to print words in a reverse lexicographical order: cat, bat, banana, band, ball, apple